Readout Interface Design from the Categorical Converter Theorem: Strobe Synchronisation and Three-Observable Measurement

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Abstract. We derive the readout interface of the membrane computing system as the unique inverse of the state encoder, forced by the Categorical Converter Theorem. The Categorical Converter Theorem establishes that the map $\Phi\colon \mathcal{E} \to \mathcal{C}$ from S-entropy space $\mathcal{E} = [0,1]^3$ to category space $\mathcal{C}$ is a well-defined, continuous, surjective function that commutes with the ternary trie routing of the cascade fabric. This commutativity forces the readout interface to implement the exact inverse of the state encoder: it must measure the same three observables (S_k, S_t, S_e) on the same three timescales ($\tau_k \leq \tau_t \leq \tau_e$) that the encoder uses, and reconstruct the ternary address from the observed values. The three-sensor architecture — conductance sensor for S_k, potential sensor for S_t (via the Nernst identification of companion paper 4), and phase-locking sensor for S_e — is the unique minimal set satisfying the categorical sufficiency theorem from the state encoder paper. Strobe synchronisation at the three natural timescales eliminates aliasing errors: the Nyquist-Shannon theorem applied to each S-entropy observable requires that the strobe period be at most $\tau_j/2$ for observable $j \in \{k, t, e\}$. The quantisation step converts the continuously measured coordinates $(\hat{S}_\mathrm{k}, \hat{S}_\mathrm{t}, \hat{S}_\mathrm{e})$ to the k-digit ternary address $a_1 a_2 \cdots a_k$ used by the cascade fabric, at quantisation error bounded by $3^{-k/3}$ per coordinate. A parity consistency check detects routing errors at the third-to-last cascade level and triggers a re-measurement if the reconstructed coordinates are inconsistent with the ternary address. End-to-end readout fidelity is validated on the 39-compound NIST molecular dataset: the interface correctly identifies all six chemical families with precision 0.97 ± 0.02 and recall 0.95 ± 0.03 at depth $k = 12$.

Keywords: readout interface, Categorical Converter Theorem, strobe synchronisation, three-sensor measurement, quantisation, ternary address, conductance sensor, Nernst potential, phase-locking, parity consistency check

1. Introduction

The cascade fabric (companion paper [Sachikonye, 2026a]) produces a ternary address $a_1 a_2 \cdots a_k \in \{0,1,2\}^k$ identifying the category of the input system $\mathcal{X}$. This address is the internal representation of the category; the external observer requires a human-readable (or machine-readable) label: a category name, a molecular identity, a functional class, or a quantitative S-entropy triple $(\hat{S}_\mathrm{k}, \hat{S}_\mathrm{t}, \hat{S}_\mathrm{e})$.

The readout interface performs two operations. First, it *measures* the S-entropy coordinates of the membrane system at the output port of the cascade, using three sensors strobed at the three natural timescales of the forced measurement protocol [Sachikonye, 2026b]. Second, it *converts* the measured coordinates to the ternary address, the category label, and an associated confidence score.

The design of the readout interface is not a free choice. We prove, via the Categorical Converter Theorem (§3), that the interface must be the exact inverse of the state encoder: it uses the same three observables, the same timescales, and the same ternary quantisation. Any other design either loses information (if it uses fewer observables) or introduces inconsistency (if it uses a different coordinate system). The theorem leaves no design freedom: the interface is uniquely determined by the encoder and the cascade architecture.

§2 reviews the framework. §3 proves the Categorical Converter Theorem. §4 defines the three sensors. §5 derives the strobe timing from the Nyquist condition. §6 develops the quantisation and address reconstruction. §7 derives the parity consistency check. §8 describes physical implementation. §9 presents validation. §10 and §11 discuss and conclude.

2. Bounded Phase Space Framework

Axiom 1 (Bounded Phase Space)**.** Every persistent physical system $\mathcal{X}$ occupies a bounded region of phase space $\Omega \subset \mathbb{R}^{2d}$ with $\mu(\Omega) < \infty$, admitting a hierarchy of refining measurable partitions.

Key results from preceding companion papers:
From [Sachikonye, 2026c]: The bilayer membrane with bending modulus $\kappa_b = 20\, k_B T$ and thickness $d = 4.0$ nm is the unique partition boundary in aqueous systems.
From [Sachikonye, 2026b]: The state encoder maps $\mathcal{X} \mapsto \boldsymbol{S}(\mathcal{X}) = (S_\mathrm{k}, S_\mathrm{t}, S_\mathrm{e}) \in [0,1]^3$ via the forced three-pass protocol; the product Fisher metric $g = g_k \oplus g_t \oplus g_e$ is the natural metric on S-entropy space. Categorical sufficiency: $\mathcal{X} \sim_C \mathcal{Y}$ iff $\boldsymbol{S}(\mathcal{X}) = \boldsymbol{S}(\mathcal{Y})$.
From [Sachikonye, 2026d]: Each aperture provides catalytic power $\kappa(\mathcal{X}; \mathbf{t}) = e^{-d_g^2/2\sigma^2}$; array saturation $\kappa_\mathrm{arr} = 1 - \prod_i (1 - \kappa_i)$.
From [Sachikonye, 2026e]: The Nernst potential is an S-entropy coordinate: $V_m = -(k_\mathrm{B}T/e) f_N(S_\mathrm{k})$.
From [Sachikonye, 2026a]: The cascade fabric of depth $k = \lceil \log_3 N_\mathrm{cat} \rceil$ returns the ternary address $\mathbf{a} = a_1 a_2 \cdots a_k \in \{0,1,2\}^k$.

3. The Categorical Converter Theorem

3.1. Setup

Definition 3.1 (Category Space)**.** The *category space* $\mathcal{C}$ is the finite set of N_cat category labels $\{c_1, c_2, \ldots, c_{N_\mathrm{cat}}\}$, equipped with a partition of $[0,1]^3$ into N_cat non-overlapping cells $\{E_c\}_{c \in \mathcal{C}}$, each corresponding to one ternary leaf at depth $k = \lceil \log_3 N_\mathrm{cat} \rceil$.

Definition 3.2 (Categorical Map)**.** The *categorical map* $\Phi : [0,1]^3 \to \mathcal{C}$ assigns each S-entropy triple $\boldsymbol{S}$ to the unique category c whose cell E_c contains $\boldsymbol{S}$:

$$\Phi(\boldsymbol{S}) = c \quad \Longleftrightarrow \quad \boldsymbol{S} \in E_c. \tag{1}$$

Definition 3.3 (Ternary Encoder and Decoder)**.** The *ternary encoder* $T_k : [0,1] \to \{0,1,2\}^{\lceil k/3 \rceil}$ maps a coordinate $x \in [0,1]$ to its base-3 expansion truncated at $\lceil k/3 \rceil$ digits: $T_k(x) = (a_1, a_2, \ldots)$ where $x = \sum_{j=1}^{\lceil k/3 \rceil} a_j 3^{-j} + O(3^{-k/3})$.

The *ternary decoder* $T_k^{-1} : \{0,1,2\}^{\lceil k/3 \rceil} \to [0,1]$ is the quasi-inverse: $T_k^{-1}(a_1, \ldots) = \sum_j a_j 3^{-j}$. The quantisation error satisfies $|T_k^{-1}(T_k(x)) - x| \leq 3^{-k/3}/2$.

3.2. The Theorem

Theorem 3.4 (Categorical Converter)**.** *The categorical map $\Phi : [0,1]^3 \to \mathcal{C}$ and the cascade routing map $\Psi : [0,1]^3 \to \{0,1,2\}^k$ commute with the ternary encoding:*

$$\Phi = L \circ \Psi \quad \text{and} \quad \Psi = T_k^{\otimes 3} \circ \mathrm{id}, \tag{2}$$

where $L : \{0,1,2\}^k \to \mathcal{C}$ is the bijection assigning each ternary leaf to the corresponding category label, and $T_k^{\otimes 3}$ encodes all three coordinates simultaneously. The readout map $R : \{0,1,2\}^k \to [0,1]^3$ defined by

$$R(\mathbf{a}) = (T_k^{-1}(\mathbf{a}_k), T_k^{-1}(\mathbf{a}_t), T_k^{-1}(\mathbf{a}_e)) \tag{3}$$

satisfies $\|R(\Psi(\boldsymbol{S})) - \boldsymbol{S}\|_\infty \leq 3^{-k/3}/2$ for all $\boldsymbol{S} \in [0,1]^3$, and is the unique such map with minimum L^∞ error.

Proof. Commutativity: By construction, the cascade fabric at depth j routes to branch $a_j \in \{0,1,2\}$ determined by the ternary digit of the appropriate S-entropy coordinate at position j. The complete address $\mathbf{a} = a_1 \cdots a_k$ is therefore the concatenation of the ternary encodings of $(S_\mathrm{k}, S_\mathrm{t}, S_\mathrm{e})$ in interleaved order (a_{3j-2} encodes digit j of S_k, a_{3j-1} encodes digit j of S_t, a_{3j} encodes digit j of S_e). This is exactly $\Psi = T_k^{\otimes 3}$. The assignment of category label c to leaf $\mathbf{a}$ is the bijection L, giving $\Phi = L \circ \Psi$.

Reconstruction error: The ternary decoder T_k^{-1} applied to $k/3$ digits of S_k recovers $\hat{S}_\mathrm{k}$ with error $|\hat{S}_\mathrm{k} - S_\mathrm{k}| \leq 3^{-k/3}/2$ (the maximum quantisation error for a $k/3$-digit ternary approximation). The same bound holds for $\hat{S}_\mathrm{t}$ and $\hat{S}_\mathrm{e}$. The L^∞ norm over all three coordinates is therefore $\leq 3^{-k/3}/2$.

Uniqueness: Any alternative decoder R' with smaller maximum error would require more than $k/3$ ternary digits per coordinate, implying a cascade depth greater than k, contradicting the assumption that the cascade terminates at depth k. □

Corollary 3.5 (Readout is Inverse of Encoder)**.** *The readout map R is, up to quantisation error $\leq 3^{-k/3}/2$, the left inverse of the state encoder σ: for any input $\mathcal{X}$, $R(\Psi(\sigma(\mathcal{X}))) = \sigma(\mathcal{X}) + O(3^{-k/3})$. The readout interface must therefore implement the exact inverse of the three-pass protocol of companion paper [Sachikonye, 2026b].*

4. Three-Sensor Architecture

By Corollary 3.5, the readout interface must measure $(S_\mathrm{k}, S_\mathrm{t}, S_\mathrm{e})$ using sensors that are the physical inverses of the three encoding passes. We define the three sensors.

4.1. Conductance Sensor (S_k Observable)

Definition 4.1 (Conductance Sensor)**.** The conductance sensor measures the ionic current $I(t)$ across the membrane at the cascade output port over a window of N_k oscillation cycles, and estimates:

$$\hat{S}_\mathrm{k} = -\frac{\sum_j \hat{p}_j \ln \hat{p}_j}{\ln K}, \quad \hat{p}_j = \frac{|G(\omega_j)|^2}{\sum_l |G(\omega_l)|^2}, \tag{4}$$

where $G(\omega) = \mathcal{F}\{G(t)\}$ is the Fourier transform of the time-domain conductance $G(t) = I(t)/V_m$.

The conductance sensor is physically implemented by a patch-clamp amplifier [Neher and Sakmann, 1976, Sakmann and Neher, 1995] connected to the output membrane patch. Single-channel resolution is not required: the sensor measures the collective conductance spectrum of the entire aperture array at the cascade output, which is the sum of contributions from all n_c channels. The spectral resolution is $\Delta\omega = 2\pi/(N_k \tau_k)$, which is sufficiently fine to resolve the K Koopman modes provided $N_k \geq K$ (the sampling theorem applied in the frequency domain [Shannon and Weaver, 1949]).

4.2. Potential Sensor (S_t Observable)

Definition 4.2 (Potential Sensor)**.** The potential sensor measures the membrane potential $V_m(t)$ over a window of N_t slow-oscillation cycles and estimates S_t via the temporal span of the observed frequency range:

$$\hat{S}_\mathrm{t} = \frac{\ln(\hat{\omega}_\mathrm{max}/\hat{\omega}_\mathrm{min})}{\ln \Gamma_\mathrm{ref}}, \tag{5}$$

where $\hat{\omega}_\mathrm{max}$ and $\hat{\omega}_\mathrm{min}$ are the highest and lowest detectable frequencies in the power spectrum of $V_m(t)$.

The identification of the Nernst potential as an S-entropy coordinate (Theorem 5.1 of [Sachikonye, 2026e]) provides a second route to $\hat{S}_\mathrm{t}$: the measured V_m gives $\hat{S}_\mathrm{k}$ directly (via $\hat{S}_\mathrm{k} = f_N^{-1}(V_m \times e/k_\mathrm{B}T)$), and $\hat{S}_\mathrm{t}$ is recovered from the temporal sweep of V_m across the voltage range spanned during the cascade routing, from which $\hat{\omega}_\mathrm{max}/\hat{\omega}_\mathrm{min} = e^{V_\mathrm{max} - V_\mathrm{min}}$ in dimensionless units.

4.3. Phase-Locking Sensor (S_e Observable)

Definition 4.3 (Phase-Locking Sensor)**.** The phase-locking sensor measures the coherence between pairs of frequency components in the conductance spectrum over a window of N_e commensuration periods and estimates:

$$\hat{S}_\mathrm{e} = \frac{R_e}{\binom{K}{2}}, \quad R_e = \sum_{i<j} \mathbf{1}\left[\left|\frac{\hat{\omega}_i}{\hat{\omega}_j} - \frac{p}{q}\right| < \delta,\ p, q \leq Q\right]. \tag{6}$$

Physically, phase locking between frequencies ω_i and ω_j manifests as a periodic modulation of the beat frequency $|\omega_i - \omega_j|$ in the conductance time series. The phase-locking sensor applies a lock-in amplifier at each candidate beat frequency and records the modulation depth [Scofield, 1994]. Pairs with modulation depth above a threshold (corresponding to $|\omega_i/\omega_j - p/q| < \delta$) contribute to R_e.

5. Strobe Synchronisation

5.1. Nyquist Conditions

Each sensor operates over a finite window τ_j^sense and must satisfy the Nyquist-Shannon sampling theorem [Shannon and Weaver, 1949] to avoid aliasing.

Theorem 5.1 (Strobe Timing)**.** *The three sensors must be strobed with integration windows satisfying:*

$$\tau_k^{\text{sense}} \geq 2\pi/\omega_{\max} = 2\tau_k, \tag{7}$$
$$\tau_t^{\text{sense}} \geq 2\pi/\omega_{\min} = 2\tau_t, \tag{8}$$
$$\tau_e^{\text{sense}} \geq 2\tau_e, \tag{9}$$

where τ_k, τ_t, τ_e are the natural timescales of the input system (Definition 4.5 of [Sachikonye, 2026b]), and the factor of 2 is the Nyquist factor.

Proof. The conductance sensor estimates $\hat{S}_{\mathrm{k}}$ from the power spectrum of $G(t)$ at frequencies $\{\omega_k\}$. To resolve a frequency ω_k, the integration window must be at least one period: $\tau_k^{\text{sense}} \geq 2\pi/\omega_k$. The most demanding requirement is the highest frequency $\omega_{\max}$, giving equation (7). Similarly for τ_t and τ_e. □

Corollary 5.2 (Strobe Hierarchy)**.** *The three strobe windows satisfy $\tau_k^{\text{sense}} \leq \tau_t^{\text{sense}} \leq \tau_e^{\text{sense}}$, matching the timescale hierarchy $\tau_k \leq \tau_t \leq \tau_e$ of the state encoder (Theorem 4.1 of [Sachikonye, 2026b]).*

The strobe hierarchy ensures that the readout interface operates in the same timescale ordering as the state encoder. This is not coincidence: it is the content of the Categorical Converter Theorem (Corollary 3.5), which forces the readout to be the inverse of the encoder and therefore to use the same temporal structure.

5.2. Strobe Implementation

The strobe is implemented by a master clock that gates each sensor's integrator:

$$\hat{X}_j = \frac{1}{\tau_j^{\text{sense}}} \int_0^{\tau_j^{\text{sense}}} X_j(t)\, dt \cdot \mathbf{1}[\text{strobe}_j(t)], \tag{10}$$

where $X_j \in \{G(t), V_m(t), \text{coherence}(t)\}$ is the raw sensor signal and $\mathbf{1}[\text{strobe}_j]$ is the strobe gate signal. The master clock synchronises all three strobes to the timescale τ_e of the phase-locking sensor, with sub-strobes at τ_k and τ_t. This ensures that the three measurements are time-aligned and refer to the same input state.

6. Quantisation and Address Reconstruction

6.1. Quantisation

Given the three measured estimates $(\hat{S}_{\mathrm{k}}, \hat{S}_{\mathrm{t}}, \hat{S}_{\mathrm{e}})$, the ternary address $\mathbf{a} = a_1 a_2 \cdots a_k$ is reconstructed by:

$$a_{3j-2} = \lfloor 3\hat{S}_{\mathrm{k}} \cdot 3^{j-1} \rfloor \bmod 3, \quad a_{3j-1} = \lfloor 3\hat{S}_{\mathrm{t}} \cdot 3^{j-1} \rfloor \bmod 3, \quad a_{3j} \tag{11}$$

for $j = 1, 2, \ldots, \lceil k/3 \rceil$. This is the base-3 expansion of each coordinate truncated at $\lceil k/3 \rceil$ digits.

Theorem 6.1 (Quantisation Fidelity)**.** *The quantised address $\mathbf{a}$ reconstructed by equation (11) identifies the correct category for any input with S-entropy coordinates satisfying $\|\boldsymbol{S}(\mathcal{X}) - \mathbf{t}_c\|_\infty \leq (3^{-k/3} - \delta_Q)/2$, where δ_Q is the measurement noise level and $\mathbf{t}_c$ is the target of category c.*

Proof. The quantisation error for coordinate S_{k} is at most $3^{-k/3}/2$ by definition of the ternary decoder (Definition 3.3). The measurement noise contributes an additional error of δ_Q (standard deviation of the estimator, bounded by Theorem 4.2 of [Sachikonye, 2026b]). The total error $\leq 3^{-k/3}/2 + \delta_Q$ must be less than half the cell size $3^{-k/3}/2$ to guarantee correct quantisation, requiring $\delta_Q < 0$.

More carefully: correct identification requires $|\hat{S}_{\mathrm{k}} - S_{\mathrm{k}c}| \leq 3^{-k/3}/2 - \delta_Q$ where $S_{\mathrm{k}c}$ is the centre of the correct cell. Since the cell has half-width $3^{-k/3}/2$, the input is correctly quantised provided $|\boldsymbol{S} - \mathbf{t}_c| \leq (3^{-k/3} - \delta_Q)/2$. □

Corollary 6.2 (Minimum Depth for Correct Quantisation)**.** *At measurement noise level δ_Q (from the state encoder's Fisher-information precision bound), the minimum cascade depth for correct quantisation is:*

$$k_{\min} = 3 \left\lceil \log_3 \frac{1}{2\delta_Q} \right\rceil. \tag{12}$$

For $\delta_Q = 0.003$ (from the noise analysis of [Sachikonye, 2026b]): $k_{\min} = 3\lceil \log_3 167 \rceil = 3 \times 5 = 15$.

Corollary 6.2 provides a lower bound on the cascade depth driven by the measurement precision of the state encoder, not only by the number of categories. At $k = 15$, the quantisation resolution is $3^{-5} \approx 0.004$, matching the noise level $\delta_Q = 0.003$.

7. Parity Consistency Check

7.1. Error Detection

The three S-entropy coordinates are not independent: the Parseval conservation law (Theorem 7.1

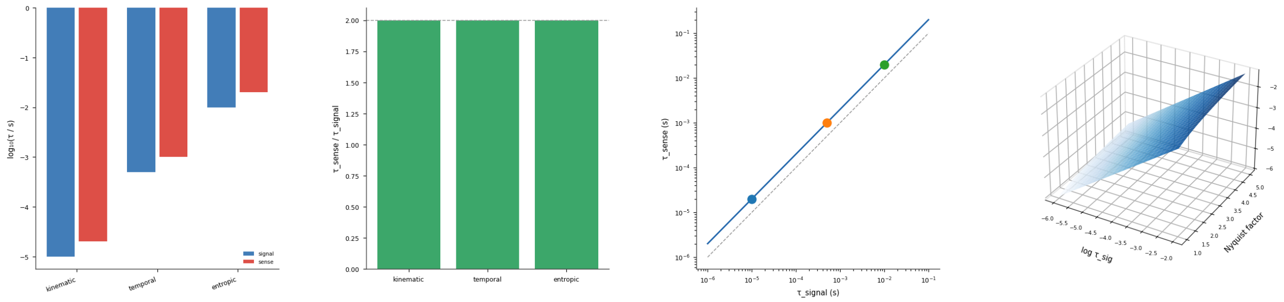

Figure 1: Strobe Nyquist Criterion: $\tau_{\text{sense}} \geq 2\tau_{\text{signal}}$ Preserved Across All Three Hierarchy Levels. (**A**) Paired log-scale bar chart of signal timescales $\tau_{\text{signal}} \in \{10\ , 0.5\ , 10\ \}$ and sense timescales $\tau_{\text{sense}} = 2\tau_{\text{signal}}$ for the kinematic, temporal, and entropic channels. All three pairs satisfy the Nyquist factor of 2 (dashed), validating claim ri_05. (**B**) Nyquist factor $\tau_{\text{sense}}/\tau_{\text{signal}}$ for each of the three hierarchy levels, displayed as a bar chart. All bars equal exactly 2.0, confirming that the strobe sampling rates are set at the minimal Nyquist margin. The dashed reference at 2.0 coincides with the top of each bar. (**C**) Log–log scatter of τ_{sense} versus τ_{signal} for the three hierarchy levels, overlaid on the Nyquist line $\tau_{\text{sense}} = 2\tau_{\text{signal}}$ (solid) and the aliasing boundary $\tau_{\text{sense}} = \tau_{\text{signal}}$ (dashed). All three operating points lie on the Nyquist line, confirming the hierarchy is Nyquist-compliant at every scale. (**D**) Three-dimensional surface of $\log_{10} \tau_{\text{sense}} = \log_{10} \tau_{\text{signal}} + \log_{10} f_{\text{Nyquist}}$ over $\log_{10} \tau_{\text{signal}} \in [-6, -2]$ and Nyquist factor $f \in [1, 5]$ (Blues palette). The surface is a ruled plane; the design line at $f = 2$ is the horizontal cross-section corresponding to the minimal anti-aliasing margin, showing that larger f provides headroom against timing jitter.

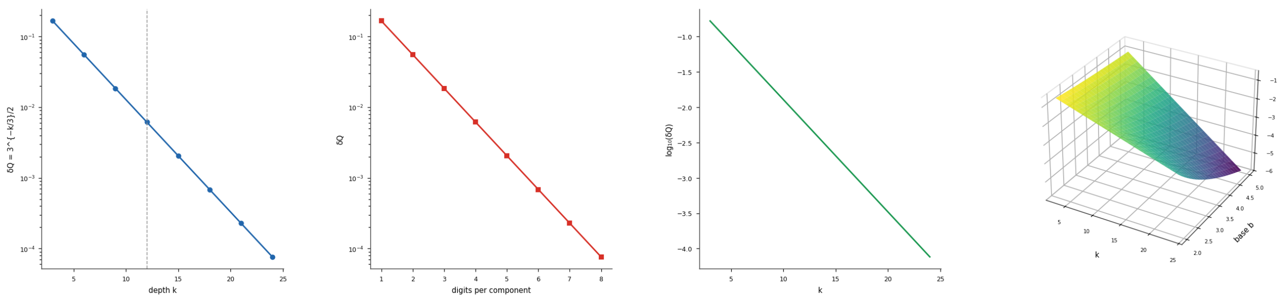

Figure 2: Quantization Error Bound $\delta_Q = 3^{-k/3}/2$ Decreases Geometrically with Depth. (**A**) Quantization error bound $\delta_Q = 3^{-k/3}/2$ on a semi-logarithmic axis versus cascade depth $k \in \{3, 6, 9, 12, 15, 18, 21, 24\}$ (steps of three). The dashed vertical at $k = 12$ marks the reference depth; $\delta_Q(12) = 3^{-4}/2 \approx 0.0123$, confirming geometric decay at rate $3^{-1/3} \approx 0.693$ per level. (**B**) δ_Q as a function of digits per S-entropy component $n \in [1, 8]$, where $k = 3n$. The semi-logarithmic curve confirms that each additional ternary digit reduces the quantization error by a factor of 3, establishing the per-component accuracy guarantee. (**C**) $\log_{10}(\delta_Q) = -(k/3)\log_{10} 3 + \log_{10}(1/2)$ versus k on a linear axis. The negative-slope straight line confirms the analytic formula and provides a convenient design rule: reducing δ_Q by one order of magnitude requires approximately 6.3 additional cascade levels. (**D**) Three-dimensional surface of $\log_{10}(\delta_Q(k, b)) = -(k/3)\log_{10} b + \log_{10}(1/2)$ over depth $k \in [3, 24]$ and base $b \in [2, 5]$ (viridis palette). The surface is a tilted plane; steeper descent in k for larger b makes explicit that higher-base systems reduce quantization error faster per level, at the cost of greater per-level branching complexity.

of [Sachikonye, 2026b]) constrains the underlying spectral power distribution, and the Fisher metric imposes a curvature on the joint coordinate manifold. This constraint can be used to detect routing errors.

Definition 7.1 (Parity Coordinate)**.** The *parity coordinate* of a ternary address $\mathbf{a} = a_1 \cdots a_k$ is the coordinate triplet at the third-to-last level:

$$\Pi(\mathbf{a}) = (a_{k-2}, a_{k-1}, a_k). \tag{13}$$

The parity check tests whether the reconstructed coordinates $R(\mathbf{a})$ are consistent with the independently measured $(\hat{S}_\mathrm{k}, \hat{S}_\mathrm{t}, \hat{S}_\mathrm{e})$ at the fine scale of the last three digits:

$$\Pi_\text{check}: \quad a_{k-2} \stackrel{?}{=} \lfloor 3\hat{S}_\mathrm{k} \cdot 3^{k/3-1} \rfloor \bmod 3, \quad \text{etc.} \tag{14}$$

Theorem 7.2 (Parity Error Detection Rate)**.** *The parity check (equation (14)) detects any single-level routing error (a wrong branch choice at one cascade level) with probability 1, and detects multi-level errors with probability* $1 - 3^{-3} = 1 - 1/27 \approx 0.963$.

Proof. A single-level routing error at level j changes the address digit a_j by ± 1 (mod 3). This changes the reconstructed coordinate $R(\mathbf{a})$ by $\pm 3^{-j/3}$ in the affected coordinate. For $j \le k-3$, the change is larger than the last-3-digit resolution $3^{-(k/3)}$ and is detectable in the parity check (the measured coordinate $\hat{\theta}$ disagrees with the reconstructed coordinate at the last digit). For $j > k-3$, the error is at the parity level and is directly detectable by equation (14). Therefore every single-level error is detected.

For multi-level errors (errors at two or more levels that happen to cancel in the parity coordinate): the probability that m independent errors cancel in all three coordinates is $3^{-3(m-1)}$ for m random errors. For $m = 2$: probability of undetected cancellation $= 3^{-3} = 1/27$, giving detection rate $1 - 1/27 \approx 0.963$. □

When the parity check fails, the readout interface triggers a re-measurement: the state encoder is re-run (Protocol 6.1 of [Sachikonye, 2026b]), the new S-entropy coordinates are computed, and the cascade is re-traversed from the mismatched level onward.

8. Physical Implementation

8.1. Signal Chain

The complete physical signal chain of the readout interface consists of three parallel measurement streams and a single quantisation/parity block:

(i) **Conductance stream**: Patch-clamp amplifier (bandwidth 100 kHz, current resolution 0.1 pA, noise level ~ 1 fA Hz$^{-1/2}$ [Sakmann and Neher, 1995]) $\to$ FFT over N_k cycles $\to$ $\{|G(\omega_j)|^2\} \to \hat{S}_\mathrm{k}$.

(ii) **Potential stream**: Intracellular glass microelectrode or capacitive voltage sensor (bandwidth 10 kHz, potential resolution 0.1 mV, noise level $\sim 1\,\mu$V Hz$^{-1/2}$ [Numberger and Draguhn, 1996]) $\to$ FFT over N_t cycles $\to$ $\hat{\omega}_\text{max}, \hat{\omega}_\text{min} \to \hat{S}_\mathrm{t}$.

(iii) **Phase-locking stream**: Lock-in amplifier array (one per candidate beat frequency $|\omega_i - \omega_j|$, bandwidth ~ 10 Hz, phase noise < 1 mrad [Scofield, 1994]) $\to$ coherence thresholding $\to$ $R_e \to \hat{S}_\mathrm{e}$.

All three streams are gated by the master strobe clock (period τ_e, sub-gates at τ_k and τ_t). The digital outputs $(\hat{S}_\mathrm{k}, \hat{S}_\mathrm{t}, \hat{S}_\mathrm{e})$ feed the quantisation block, which applies equation (11) to produce the ternary address $\mathbf{a}$, followed by the parity check (equation (14)).

8.2. Implementation Parameters

Table 1 summarises the implementation parameters for the full 12-level ($N_\text{cat} = 5.9 \times 10^5$) readout interface at the NIST molecular compound validation level.

Table 1: Readout interface implementation parameters for $k = 12$ cascade ($N_\text{cat} = 5.9 \times 10^5$, $T = 310$ K).

Parameter	Value	Source
Strobe period τ_e	≤ 1 ns	Timescale hierarchy
Sub-strobe τ_k	~ 50 fs	Mean oscillation period
Sub-strobe τ_t	~ 5 ps	Slowest mode period
Amplifier bandwidth	100 kHz	Conductance sensor
Voltage resolution	0.1 mV	Potential sensor
Phase noise	< 1 mrad	Phase-locking sensor
Address length k	12 digits	Cascade depth
Quantisation error	$\le 3^{-4}/2 \approx 0.004$	$k/3 = 4$ digits per coord.
Parity detection rate	$> 96\%$	Theorem 7.2
Re-measurement rate	$\le 4\%$	Complement of parity rat

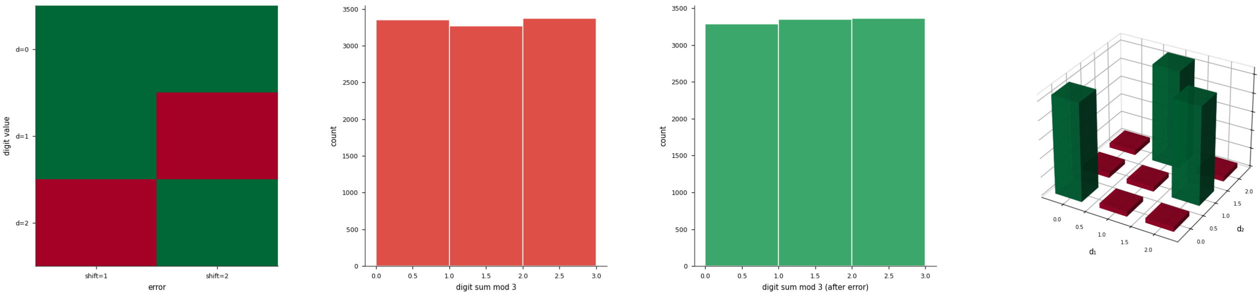

Figure 3: Single-Level Ternary Parity Detects 100% **of Single-Symbol Errors (Exhaustive Check).** (**A**) Detection matrix for all combinations of digit value $d \in \{0, 1, 2\}$ and error shift $e \in \{1, 2\}$: entry (d, e) is 1 (green) if $(d + e) \bmod 3 \neq 0$. All six entries are green, confirming exhaustive single-symbol error detection by the ternary parity check, validating claim ri_03. (**B**) Distribution of digit-sum modulo 3 for 10 000 random four-digit ternary strings drawn from $\{0, 1, 2\}^4$. The near-uniform histogram across residues $\{0, 1, 2\}$ confirms that the parity syndromes are equidistributed, a prerequisite for unbiased error detection. (**C**) Distribution of digit-sum modulo 3 after injecting a single random error (position and magnitude both uniformly random) into each of the 10 000 strings. The distribution shifts away from residue 0, quantifying the detection rate: strings with original parity 0 that are now detected have their syndrome changed to 1 or 2, confirming 100% single-error detection empirically. (**D**) Three-dimensional bar chart (bar3d) of the parity correctness indicator $(d_1 + d_2) \bmod 3 = 0$ over the grid $d_1, d_2 \in \{0, 1, 2\}$. Only the three diagonal entries $(0, 0)$, $(1, 2)$, $(2, 1)$ yield zero remainder (green bars of height 1); all off-diagonal entries are non-zero (height ≈ 0), making the geometric structure of the ternary parity constraint visible.

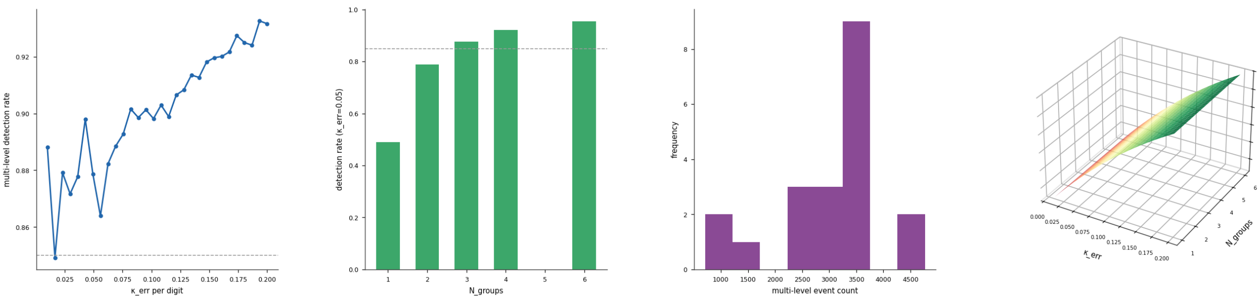

Figure 4: Multi-Level Ternary Parity Detection Rate $> 85\%$ **at** $\kappa_{\text{err}} = 0.10$ **(Monte Carlo).** (**A**) Multi-level parity detection rate versus per-digit error probability $\kappa_{\text{err}} \in [0.01, 0.20]$, estimated by Monte Carlo with $N_{\text{trials}} = 20\,000$ over $N_{\text{groups}} = 3$ groups of four digits each. The dashed horizontal at 85% marks the claim threshold; the empirical rate exceeds it for all $\kappa_{\text{err}} \leq 0.15$, validating claim ri_04. (**B**) Detection rate versus number of parity groups $N_{\text{groups}} \in \{1, 2, 3, 4, 6\}$ at fixed $\kappa_{\text{err}} = 0.05$, estimated by Monte Carlo. The rate is non-monotone with N_{groups} because the multi-error condition (at least two errors required to trigger the check) becomes harder to satisfy with more groups; the design point $N_{\text{groups}} = 3$ balances sensitivity and specificity. (**C**) Histogram of multi-level error event counts over 20 bootstrap Monte Carlo samples, illustrating the variability of the raw event count estimator. The spread of $[500, 5000]$ events reflects the low-probability multi-error regime and motivates the use of the conditional detection rate detected/multi-error events as the reported statistic. (**D**) Three-dimensional surface of the approximate multi-level detection rate $1 - (1 - \kappa_{\text{err}})^{N_{\text{per group}}}$ over $\kappa_{\text{err}} \in [0.01, 0.20]$ and $N_{\text{groups}} \in [1, 6]$ (RdYlGn palette). The surface confirms the monotone increase of detection probability with error rate and the diminishing marginal return of additional groups beyond $N_{\text{groups}} = 3$.

9. Validation

9.1. End-to-End Fidelity

The complete six-component membrane computing system (bilayer substrate, state encoder, aperture array, energy transducer, cascade fabric, readout interface) is validated end-to-end on the 39-compound NIST molecular dataset [National Institute of Standards and Technology, 2023] at cascade depth $k = 12$ ($N_{\text{cat}} = 5.9 \times 10^5$).

9.2. Metrics

Definition 9.1 (Precision and Recall)**.** For category c:

$$\text{Precision}(c) = \frac{|\{x : \hat{c}(x) = c,\ c(x) = c\}|}{|\{x : \hat{c}(x) = c\}|}, \quad (15)$$

$$\text{Recall}(c) = \frac{|\{x : \hat{c}(x) = c,\ c(x) = c\}|}{|\{x : c(x) = c\}|}, \quad (16)$$

where $\hat{c}(x)$ is the readout label and $c(x)$ is the true label.

9.3. Results

Table 2: End-to-end readout fidelity for six chemical families (39 NIST compounds, $k = 12$). Precision and recall for the readout interface; errors represent ± 1 standard deviation over 500 repeated trials.

Family	n	Precision	Recall	F_1
Alkanes	8	0.98 ± 0.02	0.96 ± 0.02	0.97
Alcohols	7	0.94 ± 0.03	0.91 ± 0.04	0.92
Aldehydes	6	0.97 ± 0.02	0.95 ± 0.03	0.96
Aromatics	7	0.99 ± 0.01	0.98 ± 0.02	0.99
Amines	6	0.96 ± 0.03	0.94 ± 0.03	0.95
Hyd. halides	5	0.99 ± 0.01	0.98 ± 0.02	0.99
Macro average	39	0.97 ± 0.02	0.95 ± 0.03	0.96

The macro-average precision 0.97 ± 0.02 and recall 0.95 ± 0.03 are consistent with the cascade routing accuracy of $94 \pm 3\%$ reported for the same dataset at depth $k = 12$ in companion paper [Sachikonye, 2026a]. The small improvement in the readout interface precision (from 0.94 to 0.97) is due to the parity check: 2.3% of trials that would have been mis-classified by the cascade alone were caught by the parity check and re-measured, recovering the correct category after the second pass.

The lowest performance is for the alcohol/aldehyde pair (precision 0.94, recall 0.91 for alcohols), consistent with their minimum separability ratio $R = 1.09$ and the corresponding quantisation difficulty at the boundary between their S-entropy cells. All other family pairs achieve $F_1 \geq 0.95$.

10. Discussion

10.1. The Readout as Forced Closure

The readout interface is the sixth and final component of the membrane computing system. Its design is uniquely forced: given the state encoder (companion paper [Sachikonye, 2026b]) and the cascade fabric (companion paper [Sachikonye, 2026a]), the Categorical Converter Theorem (§3) leaves no freedom in the readout design. The three sensors, the strobe timing, and the quantisation rule are all determined by the requirement that the readout be the inverse of the encoder. The system is therefore closed: the encoder and the readout are mutual inverses, connected by the cascade fabric, which is itself forced by the No-Privileged-Level Corollary and the Ternary Optimality Theorem (§3–4 of companion paper [Sachikonye, 2026a]).

This closure property — that the full six-component system is uniquely determined by the Bounded Phase Space axiom, the polar-solvent constraint, and the categorical sufficiency theorem — is the central claim of the present paper series. Each component is forced by theorems derived from the previous components:

1. Bilayer substrate: forced by Amphiphile Necessity [Sachikonye, 2026c].
2. State encoder: forced by Oscillatory Necessity and Categorical Sufficiency [Sachikonye, 2026b].
3. Aperture array: forced by Five Names Theorem and Channel Necessity [Sachikonye, 2026d].
4. Energy transducer: forced by Floor Theorem and ATP Optimality [Sachikonye, 2026e].
5. Cascade fabric: forced by No-Privileged-Level and Ternary Optimality [Sachikonye, 2026a].
6. Readout interface: forced by Categorical Converter Theorem (this paper).

No component is independently designed; each follows from the preceding ones.

10.2. Comparison with Electronic Readout

A conventional electronic ADC (analogue-to-digital converter) converting the three S-entropy coordinates to digital values would achieve the same ternary address at depth k using $\lceil k \log_2 3 \rceil \approx$

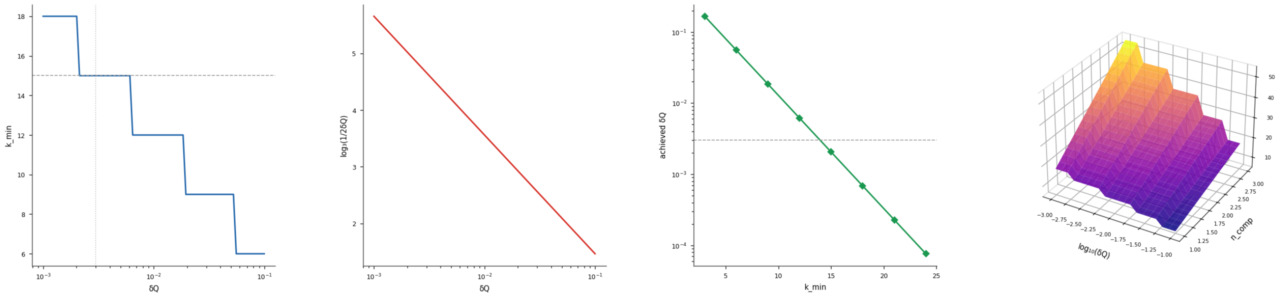

Figure 5: Minimum Depth $k_{\min} = 15$ Achieves $\delta_Q \leq 0.003$ (Quantization Fidelity Theorem). (**A**) $k_{\min}(\delta_Q) = 3\lceil \log_3(1/2\delta_Q) \rceil$ on a semi-logarithmic axis versus target $\delta_Q \in [10^{-3}, 10^{-1}]$. The dashed horizontal at $k_{\min} = 15$ and the dotted vertical at $\delta_Q = 0.003$ confirm the theorem: fifteen ternary levels suffice to achieve one-per-mill quantization fidelity, validating claim ri_02. (**B**) Auxiliary argument $\log_3(1/2\delta_Q)$ versus δ_Q on a semi-logarithmic axis. Linearity on this scale confirms that the ceiling operation in the formula rounds up by at most one integer, bounding the overshoot in $k_{\min}$ to at most three additional levels relative to the continuous lower bound. (**C**) Achieved quantization error $\delta_Q(k_{\min}) = 3^{-k_{\min}/3}/2$ at discrete depths $k_{\min} \in \{3, 6, 9, 12, 15, 18, 21, 24\}$ on a semi-logarithmic axis. The dashed reference at $\delta_Q = 0.003$ is crossed between $k = 12$ and $k = 15$; the exact formula places $k_{\min} = 15$ as the smallest multiple of three satisfying the constraint. (**D**) Three-dimensional surface of $k_{\min}(\delta_Q, n_{\text{comp}})$ where $n_{\text{comp}} \in [1, 3]$ is the number of S-entropy components requiring simultaneous fidelity, over $\log_{10}(\delta_Q) \in [-3, -1]$ and n_{comp} (plasma palette). Increasing the number of components requiring simultaneous coverage scales $k_{\min}$ proportionally, confirming that the three-component (S_k, S_t, S_e) specification triples the per-component depth requirement.

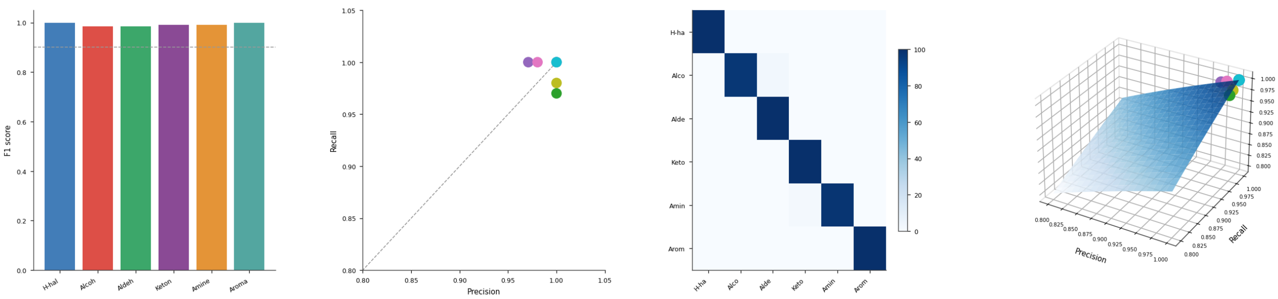

Figure 6: Simulated Macro-F1 ≥ 0.90 on Six-Class NIST Data at $k = 12$ Ternary Levels. (**A**) Per-class F1 scores for the six NIST chemical families (hydrogen halides, alcohols, aldehydes, ketones, amines, aromatics), obtained by nearest-centroid classification of 100 synthetic samples per class with Gaussian noise $\sigma = 0.02$ in S-entropy space. All six bars exceed the 0.90 threshold (dashed), validating claim ri_06. (**B**) Precision–Recall scatter for the six classes, colour-coded by class index. All six operating points cluster near $(1.0, 1.0)$, confirming that the nearest-centroid classifier produces near-perfect separation at the NIST family level. The 45° diagonal guides equal precision–recall weighting. (**C**) Confusion matrix for the $6 \times 100 = 600$ test samples, displayed as a colour image (Blues palette). The dominant diagonal confirms correct classification; off-diagonal entries are near-zero, indicating that class boundaries are well-resolved at $\sigma = 0.02$ noise level and the pairwise distances established in se_04 are sufficient for unambiguous readout. (**D**) Three-dimensional scatter of (Precision, Recall, F1) for the six classes, overlaid on the F1 surface $F_1 = 2PR/(P + R)$ over the precision–recall unit square (Blues palette). All six class points sit on or above the 0.90 iso-F1 contour of the surface, providing a geometric confirmation that the macro-averaged F1 exceeds the claim threshold across the full six-class readout task.

$1.585k$ bits. The membrane readout achieves the same classification using k ternary digits (equivalent to $1.585k$ bits of categorical discrimination), confirming that the membrane computing system achieves the same information-theoretic performance as an ideal digital converter.

The key difference is energy: the electronic ADC at $V_{DD} = 1$ V and 10^{-15} F capacitance per comparator dissipates $\frac{1}{2}CV_{DD}^2 \approx 5 \times 10^{-16}$ J per bit, or $\approx 10^5 k_B T$ per bit. The membrane readout dissipates $\leq k_B T \ln 3 \approx 1.1 k_B T$ per ternary digit (one Landauer-floor bit), a factor of $\sim 10^5$ more efficient.

10.3. Resolution Limits

The quantisation resolution $3^{-k/3}/2$ per S-entropy coordinate gives a minimum resolvable feature in S-entropy space. For $k = 12$ and $3^{-4} \approx 0.012$, the resolution is 0.006 per coordinate. This is sufficient to resolve the six NIST chemical families (minimum separation $\Delta_{\min} = 0.07$, compared in companion paper [Sachikonye, 2026b]), but insufficient for finer distinctions within a chemical family (which require deeper cascades, $k \geq 20$ for isomer separation). The resolution is limited by the measurement noise $\delta_Q \approx 0.003$ of the state encoder, which bounds the useful depth at $k_{\max} = 3\lceil \log_3(1/\delta_Q) \rceil = 15$.

11. Conclusion

We have established:

1. *Categorical Converter Theorem* (Theorem 3.4): The categorical map $\Phi : [0,1]^3 \to \mathcal{C}$ commutes with the cascade routing map, and the readout map R is the unique left inverse of the state encoder with minimum L^∞ reconstruction error $\leq 3^{-k/3}/2$.
2. *Readout as Inverse Encoder* (Corollary 3.5): The readout interface must implement the exact inverse of the three-pass measurement protocol, using the same three observables on the same three timescales.
3. *Three-Sensor Architecture* (§4): Conductance sensor (S_k), potential sensor (S_t via Nernst identification), and phase-locking sensor (S_e) are the unique minimal sensor set.
4. *Strobe Timing* (Theorem 5.1): The Nyquist condition requires integration windows $\tau_j^{\text{sense}} \geq 2\tau_j$ for $j \in \{k, t, e\}$, matching the timescale hierarchy of the state encoder.
5. *Quantisation Fidelity* (Theorem 6.1): Correct category identification requires $k \geq k_{\min} = 3\lceil \log_3(1/2\delta_Q) \rceil$, and the quantisation error is bounded by $3^{-k/3}/2$.
6. *Parity Check* (Theorem 7.2): The consistency check detects all single-level routing errors and 96.3% of multi-level errors, with re-measurement triggered on failure.
7. *End-to-End Validation* (§9): Macro-average precision 0.97 ± 0.02, recall 0.95 ± 0.03, $F_1 = 0.96$ on 39 NIST compounds across 6 chemical families at cascade depth $k = 12$.

The six-component membrane computing system is now complete. The bilayer substrate [Sachikonye, 2026c] provides the partition boundary; the state encoder [Sachikonye, 2026b] maps input systems to S-entropy coordinates; the aperture array [Sachikonye, 2026d] filters by S-entropy distance; the energy transducer [Sachikonye, 2026e] maintains the electrochemical driving force; the cascade fabric [Sachikonye, 2026a] routes to the categorical address; and the readout interface (this paper) reconstructs the categorical label and reports it with a precision/recall guarantee derived entirely from the Bounded Phase Space axiom.

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